04/05/2026

AI TOOLS

AI -

ChatGPT

Machine learning, a branch of AI that enables computers to learn and improve at tasks by finding patterns in data, rather than following rigid, step-by-step instructions.

training

data is required

data Science.

Hallucinating

RAG - Retrieval Augmented Generation.

PROMPT ENG

large language model

1. Few-shot prompting - you give the LLM a few examples so that it understands your question.
2. Zero-shot prompting - when you ask a question without giving any example.
3. Contextual prompting - you give background info
4. Constraint-based prompting - you limit the response

07/04/2026

DATA:

* Information / collection of facts used for certain things

1. STRUCTURED DATA

- Data that is stored in a table with rows and columns.
- It is consistant
- Spreadsheet.
- Application form

organised.

Relational structure data.

2. UNSTRUCTURED DATA

- Social media
- PDF documents
- Music
- Videos
- Phone call

No consistant format

RELATIONAL DATABASE (RDB)

store

→ A storage place for structured data.
→ A database is where we store the data.

DATABASE MANAGEMENT SYSTEM (DBMS)

access

→ A system we use to access the database.
A tool that allows us to interact with the data.

access, retrieve

Summary:

- Online transactional processing OLTP
extract, transform, Load ETL

DATA WAREHOUSE: Central place where we store multiple databases.

online analytical processing OLAP

storage companies

AWS → Amazon Web Services
GCP → Google Cloud Platform
AZURE →
ORACLE →

(business) analytical users

OLTP - Online Transactional Processing Database
C R U D - Delete
Create Update ⇒ Allows basic data processing operations
Retrieve

DMS - Data Management System.
↓
SQL - Structured Query Language
- Data cleaning, data exploration,
Data Inspection, data processing, wrangling
Data aggregation, Data joining, Filters

* for summarised specific data.

09/04/2026

Primary key
Foreign key

DATA WAREHOUSE
→ Central data store
→ Houses multiple databases.

→ quantitative → easy to analyse for processing

Dimension tables	Fact tables
- A table that stores descriptive table	- Stores transactional information (numeric values to measure) - Records what happened - Answers the following: - What was sold - How much was sold - How much money was made Stores: Sale_ID, Date-ID, Product-ID, Store-ID, Quantity, Revenue.

Offers: Description of what is in the fact table.

Foreign key
Primary key

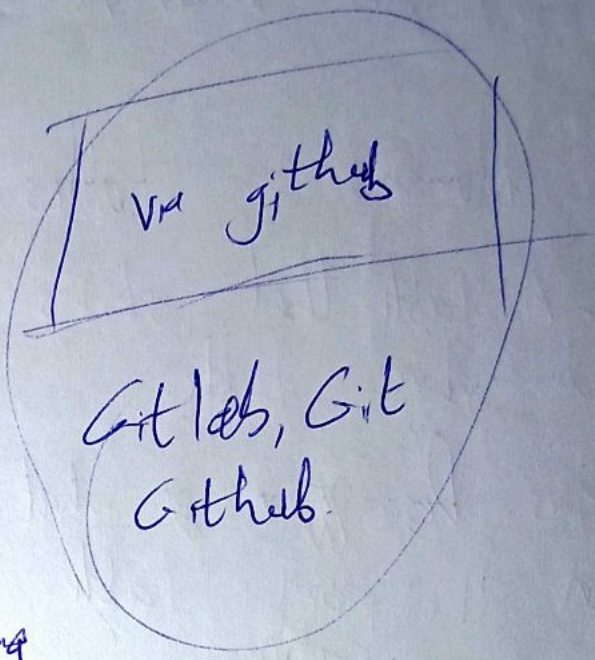
PRIMARY KEY:
A column that uniquely identifies each row in a table

FOREIGN KEY:
A column that ~~links~~ links one table to another

DATA LAKE!

What is a data lake
data mart
foreign key
primary key
Data base
Data warehouse
schema
AI
Prompt Engineering
Version control

Microsoft SQL server



13/04/2026

1. Data warehouse
2. SQL / OMS (Snowflake / Databricks / Big Query)
3. Summary data
4. Use it (Excel / Power BI)
   Machine learning

Python for functions
Pandas → PySpark

OPERATIONS:

Arithmetic operations

Operation	Description
+	Addition
−	Subtraction
/	Division
*	Multiplication

Comparison operations

Operation	Description
=	Equals to
>	Greater than
<	less than
>=	Greater than or =
<=	Less than or =

Logical Operators

Operation	Description
AND	
BETWEEN	
IN	
NOT	
OR	

ALIASES AS function.

⇒ a function used to rename / to give something a nickname

14/04/2026

SQL FUNDAMENTALS!

Select * ⇒ allows us to select all the columns in the table

1. Select statement function - allows us to select specific columns or rows in our data

2. From statement - allows us to specify the name of table we are selecting the data from.

Students_table

Student_ID	Name	Subject	Marks
101	Sipho	Geo	60
102	Mbali	Maths	45
103	Third	Maths	99

I want to know only their names and subjects.

↓

```
SELECT name, subject
FROM student_table;
```

⇒

Name	Subject
Sipho	Sipho
102	Mbali
103	Third

- We separate the column names with a comma within the select statement

3. Select Distinct - allows us to select different values in a specific column
   * Removes duplication
   * Does not repeat the same value

4. Where statement - it allows us to filter our data on specific conditions
   * It affects the rows, not the columns.
   * You filter rows based on a condition applied into a column

DATA FILTERING.

16/04/2026

LOGICAL OPERATORS

AND OPERATOR - all conditions must be met.
- You can put more than one conditions
- Join multiple conditions.

OR operator - One of the condition must be met.

IN operator - ~~allows us dat~~
=allows us to filter data by specifying multiple values in a column

* BETWEEN operator - inclusive statement
=allows us to filter the data based on a specific range in our numeric & date columns

* NOT operator - gives you the negative result
so a way of filtering by saying what will be excluded.

18/04/2026

DATA FILTERING

Give me a list of students who obtained marks above 80%

```
SELECT *
FROM Students_table
WHERE marks > 80;
```

filter here
- affects rows

* Words and dates, you put them in single quotes

all conditions must be met/satisfied

AND OPERATOR -

```
SELECT name, subject, marks
FROM Students_tabe
WHERE marks > 50 AND subject = 'math'
```

OR statement - one of the conditions must be met.

IN operator - allows us to specify multiple values in a column

IN ('value 1', 'value 2', 'value 3', ..., 'value n')



20/04/2024

AGGREGATE FUNCTIONS

1. SUM( )
2. AVG( )
3. MIN( )
4. MAX( )
5. COUNT( )

- Consolidate the data.
- Summarise the data.
- Returns a single value

= a function that performs a calculation on multiple rows in a specific column and returns a single value or a single value per group.

(1) SUM - summarises / adds up

(2) AVG

3) MIN - Gives minimum value.

4) MAX - Gives maximum value

5) COUNT - Counts how many things are available

* Count something that is unique.

~~sum~~ ID, but ~~sum~~ count it.

4. COUNT () — how many rows
b) SUM () — Total of a numeric column
- Addition of values
3. MIN () - Smallest value
4. MAX () - Largest value.
5. AVG() - Average value of a numeric column

* We count ID/Something unique

~~Filtering~~ Ordering

Sorting Statements

ORDER BY { ASC - ascending order
DESC - descending order

22/04/2026

SORTING STATEMENTS:

- It allows us to rearange our data without changing anything in that
- It does not remove or add anything, it rearranges.

1. ORDER BY statement:
   - allows us to arrange our data in 2 ways:
     1. Ascending order (ASC) - smallest to biggest
     2. Descending order (DESC) - biggest to smallest.

Syntax:

SELECT column 1, column 2, ..., column N
FROM table_name AS descriptive name/Rename a column
condition ; After aggregation a column
condition categorical column
column; ASC/DESC; - arranges the rows

LIMIT statement

- allows you to specify the number of row you want to see in your outcome

SELECT *
FROM products
LIMIT 2;

- Used to structure of the table
- inspection.

23/04/2026

HAVING STATEMENT

- It allows us to filter data that has already been aggregated

Synthax:

```
SELECT .....
agg-fund ...
FROM ....
GROUP BY
HAVING ... ;
```

- Having statement allows us to filter data that is grouped
- We group by a column that is not aggregated

WHERE statement - filters raw data
HAVING statement - filters grouped aggregated data

To begin with HAVING statement, you need to have data and group it. Then you can apply HAVING statement.

SUCCESS

1. Choose one thing and be very good at it
   * What do I need to know to be the best

28/05/2026

DATE

- A data type that shows periods of the data collection.

* Dayname (2026-05-28)
  - It extracts the day of the week based on your date column

* ~~Date~~. Monthname
  - It extracts the month name based on your date column

* Year
  - Extract the year value based on your date column

* Day
  - Extrad the day value from your date column

04/06/2026.

NULL FUNCTIONS

- NULL - an empty space
  - a missing value
  - a blank space
  - no value
- Null functions help us to deal with null values

1. IS NULL - this is a function that helps us find a row of a specific column that has a null value.

SYNTAX:

```
SELECT column....
FROM table-name
WHERE column is NULL;
```

✓ filters the rows based on the specific column

Customers

Cust-id	Name	Email
1	John	John@gmail.com
2	Mary	null
3	Peter	peter@gmail.com.

```
SELECT *
FROM customers
WHERE email is null;
```

⇒

Cust-id	Name	Email
2	Mary	null

2. NOT NULL - opposite of IS NULL.

How to replace a NULL value

If NULL (expr, default-value)

- a way to replace null values without alternative value of your choice

09/06/2026

WILD CARDS

- a way of filtering data using a specific pattern in a specific column.
- You filter data based on how words are arranged on a specific column.
    A) % TYPE - Represents zero or more characters.
    B) _ TYPE - Represents a single character
- In order to use wildcards we need to use the LIKE statement
- The LIKE statement is commonly used in the WHERE statement to filter data.

11/05/2026

CASE STATEMENT - to build logical condition.
- Allows us to build logical conditions

```
SELECT
    CASE
    WHEN condition 1 THEN result 1
    WHEN condition 2 THEN result 2
    ELSE default result
    END AS column name
    FROM table name
```

→ We write a case statement within the select statement

→ A case statement creates a new column

- We build conditions based on other columns
- A case statement is a way of building a categorical column using numerical column

31. INCREMENTAL REVENUE

= The additional ~~reneto~~ revenue generated as a direct result of a specific action or campaign

* A promotion that increases sales from R50 000 to R65 000 adds R15 000 incremental revenue.

-SOURCE: ~~Mear~~ Marketing Metrics - Paul W. Farris

32. RETURN ON INVESTMENT (ROI)

= A measure of profitability comparing net profit to the cost of investment.

* Spending R1000 and earning R1 500 gives a 50% ROI

-SOURCE: Financial Intelligence - Karen Berman.

33. RETURN ON AD SPEND (ROAS)

= Revenue earned for every unit of money spent on advertising.

* Spending R2000 on ads and earning R8000 give 4:1 ROAS

-SOURCE: Digital Marketing Analytics - Chuck Hemann.

34. CLICK-THROUGH RATE (CTR)

= The percentage of people who click on a link after seeing it.

* 50 clicks from 1000 views = 5 % CTR

-SOURCE: Internet Marketing - Dave Chaffey

35. BOUNCE RATE
= The percentage of visitors who leave a website after viewing only one page.
* A user opens a webpage and leaves without clicking anything else
-SOURCE: Web Analytics 2.0 - Avinash Kaushik.

36. A/B TESTING
= A method of comparing two versions of something to see which performs better
* Testing two different email subject lines to see which gets more opened.
-SOURCE: You Should Test That! - Chris Goward.

37. LEAD GENERATION
= The process of attracting and capturing potential customer's interest.
* Collecting email addresses through a sign-up form
-SOURCE: Inbound Marketing - Brian Halligan.

38. MARKETING ATTRIBUTION
= The process of identifying which marketing channels contribute to conversions.
* Determining whether a sale came from a Facebook ad or email campaign.
-SOURCE: Marketing Attribution - Jim Sterne.

viewing who leave a website

29. DATA PRIVACY

= refers to the protection of personal information and how it is collected, stored and used by organisations

- Data privacy is important because personal data can be sensitive, and if misused it can lead to identity theft, fraud, or loss of trust.

GDPR (GENERAL DATA PROTECTION REGULATION)

= Data protection law introduced by the European Union in 2018.

- It applies to all countries in the European Union and European Economic Area.

30. DATA ETHICS

= refers to the moral principles and guidelines that govern how data is collected, used, shared, and analysed.

EXAMPLES;

- UNAUTHARISED DATA COLLECTION

When companies collect personal data without user consent.
- This is problematic because it violates privacy rights.

- BIASED AI DECISION-MAKING

When an AI system is trained on biased data and produces unfair outcomes.
- This is problematic because it leads to discrimination and unfair treatment of certain groups

- DATA MISUSE OR SELLING PERSONAL INFORMATION

When organisations sell or share user data with third parties without clear consent.
- This is problematic because it can expose sensitive information and break trust between users and organisations

32. BIAS IN AI

= refers to systematic errors in an AI system that cause it to produce unfair or inaccurate results for certain groups of people

- An AI model learns from historical data, If that data reflects existing human prejudices or is not representive of all groups, the model will learn and repeat those mistakes or patterns. This results in decisions that are not neutral or fair

- Unfair discrimination against certain groups (race, gender, age)
- Reduced trust in AI systems
- Legal and ethical issues for companies

31. DATA GOVERNANCE

= a set of policies, processes, and standards that ensure data is managed properly

KEY COMPONENTS

- DATA QUALITY: Ensures data is accurate
- DATA SECURITY AND PRIVACY: Protects data from unauthorised access
- DATA OWNERSHIP AND ACCOUNTABILITY: Clearly defines who is responsible for managing and maintaining data.
- DATA POLICIES AND STANDARDS: Establishes rules for how data should be collected and used.
- ~~DATA RECYCLE~~
- DATA LIFECYCLE MANAGEMENT: Manages data from creation to delition.

27. AVERAGE ORDER VALUE (AOV)
= The average amount spent per order.
* R5000 revenue from 100 orders = ~~R500~~ R50 AOV.
-SOURCE: E-commerce Analytics - Judah Phillips

28. SALES FUNNEL
= The stages a customer goes through ~~th~~ from awareness to purchase
* Awareness → Interest → Decision → Purchase.
-SOURCE: Dot Com Secrets - Russell Brunson.

29. MARKETING CAMPAIGN
= A planned set of marketing activities aimed at promoting a product or service
* A company running social media ads and email promotions for a new product launch
-SOURCE: Marketing Management - Philip Kotler

30. POST-CAMPAIGN ANALYSIS
= The evaluation of a marketing campaign's performance after it ends
* Reviewing sales and engagement data to see if the campaign met its goals
-SOURCE: Marketing Analytics - Wayne L. Winston.

23. CUSTOMER LIFETIME VALUE (CLV)

= The total revenue a business expects from a customer over their relationship

* A customer spending Ksoo per year for 5 years = K2500 CLV

-SOURCE: Customer Centricity - Peter Fader

24. CHURN RATE

= The percentage of customers who stop using a product or service over time.

* Losing 20 out of 200 customers in a month = 10% churn rate

-SOURCE: Subscribed - Tien Tzuo

25. CUSTOMER RETENTION RATE (CRR)

= Measures the percentage of customers a company retains during a specific timeframe.

* Netflix subscription

-SOURCE: Netflix Annual Report.

26. NET PROMOTER SCORE (NPS)

= A metric that measures customer loyalty based on likelihood to recommend

* Customers rating 9-10 are promoters, ~~boo~~ boosting NPS

-SOURCE: The Ultimate Question 2.0 - Fred Reichheld.

19. BENCHMARK

= A standard or point of reference used to compare performance

* Comparing your company's sales growth to industry averages

-SOURCES: Benchmarking - Robert C. Camp

20. ~~Segmentation~~ SEGMANTIATION

= The process of dividing data or customers into groups based on shared characteristics

* Grouping customers by age, location or buying behaviour.

-SOURCE: Marketing Management - Philip Kotler

21. CONVERSION RATE

= The percentage of users who take a desired action

* 100 purchases from 1000 visitors = 10 % conversion rate

-SOURCE: Digital Marketing Analytics - Chuck Hemann.

22. CUSTOMER ACQUISITION COST (CAC)

= The total cost of acquiring a new customer

* Spending R10 000 to gain 100 customers = R100 CAC

-SOURCE: Lean Analytics - Alistair Croll

10. DATA LITERACY
= The ability to read, understand, and use data effectively
* A manager interpreting charts to make business decisions
- SOURCE: Be Data Literate - Jordan Morrow

11. STRUCTURED DATA
= Data that is organised in a fixed format (rows and columns)
* Customer details stored in a spreadsheet
- SOURCE: Database System Concepts - Abraham Silberschatz
- W3SCHOOLS

12. UNSTRUCTURED DATA
= Data that has no format
* Emails, videos, pictures
- SOURCE: W3SCHOOLS

13. DATA WAREHOUSE
= A centralised system that stores large amounts of data.
* A company storing years of sales data.
- SOURCE: W3SCHOOLS

14. DATA MINING
= The process of discovering patterns and relationships in large datasets
* Identifying buying patterns in customer purchase data.
- SOURCE: W3SCHOOLS

15. DATA VISUALISATION

= The presentation of data in graphical formats like charts or graphs

* A bar chart showing monthly revenue

-SOURCE: Storytelling with Data - Cole Nussbaumer Knaflic

16. DASHBOARD

= A visual display of key data and metrics in one place for quick monitoring

* A screen showing sales, profit, and customer numbers in real time.

-SOURCE: Information Dashboard Design - Stephen Few.

17. METRIC

= A measurable value used to track performance or progress

* Monthly revenue or number of customers

-SOURCE: Key Performance Indicators - David Parmenter.

18. KEY PERFORMANCE ~~INDITOR~~ INDICATOR (KPI)

= A specific metric used to evaluate how well an organisation is achieving its goals.

* ~~Custorer~~ Customer retention rate for a subscription business

-SOURCE: Key Performance Indicators - David Parmenter

5. BIG DATA
= Large and complex datasets that require advanced tools to process and analyse.
* Social media platforms handling millions of user posts daily
-SOURCE: Big Data: A Revolution - Viktor Mayer-Schönberger

6. DESCRIPTIVE ANAYTICS
= Analysis that explains what has happened in the past using data
* A monthly report showing total sales.
-SOURCE: Data Science for Business - Foster Provost

7. DIAGNOSTIC ANALYTICS
= Analysis that explains why something happened
* Investigating why sales dropped in a specific month
-SOURCE: Data Science for Business - Foster Provost

8. PREDICTIVE ANALYSIS
= Analysis that uses data to forecast future outcomes.
* Predicting customer demand for next season.
-SOURCE: Predictive Analytics - Eric Siegel.

9. PRESCRIPTIVE ANALYTICS
= Analysis that suggests actions to achieve desired outcomes
* Recommending the best ~~out~~ price to maximize profit
-SOURCE: Analytics at Work - Thomas H. Davenport

1. BUSINESS ANALYTICS

= The use of data, statistical methods, and models to solve business problems and support decisions.

* A company analysing sales to decide which products to promote.

- SOURCE: Business Analytics - James R. Evans

2. DATA ANALYTICS

= The process of examining data to find patterns, trends, or any useful insights.

* Tracking website visitors to see which pages get the most traffic

- SOURCE: Business Intelligence Guidebook - Rick Sherman

3. BUSINESS INTELLIGENCE (BI)

= Technologies and tools used to collect, process, and present business data for decision-making.

- SOURCE: Business Intelligence Guidebook - Rick Sherman.

* A dashboard showing real-time company performance metrics

4. DATA-DRIVEN DECISION MAKING

= Making decisions based on data analysis rather than intuition or guesswork.

* A retailer using sales data to decide store locations

SOURCE: Competing on Analytics - Thomas H. Davenport

ROLE	MAIN FOCUS	KEY SKILLS	COMMON TOOLS
Data analyst	-Analysing historical data to find insights and support ~~dec~~ business decisions	Data cleaning, visualisation, reporting, SQL, basic statistics	Excel, SQL, Power BI, Tableau, Python
Data scientist	Building predictive models and advanced analytical skills	Machine learning, statistics, data modeling, programming,	Python, R, Scikit-learn, TensorFlow
Data engineer	Designing and maintaining data pipelines and infrastructure	Data pipelines, ETL processes, database systems, cloud platforms	Apache Spark, Hadoop, SQL, cloud services
AI engineer	Developing AI systems and deploying machine learning models into production	Deep learning, neural networks, model deployment.	TensorFlow, ~~Python~~ PyTorch, Python, cloud AI platforms

3.4 BUSINESS INTELLIGENCE (BI)

= refers to the processes, technologies, and tools used to collect, analyse, and visualise business data in order to support better decision-making

- POWER BI: Microsoft tool used to create interactive dashboards and reports
- TABLEAU: Allows users to build interactive graphs, charts and dashboards

* BI tools combine data from multiple sources and present it in a clear, visual format. This allows decision making to be easier. Businesses ~~often~~ make faster, more accurate, and evidence-based decisions

5.

STRUCTURED DATA	SEMI-STRUCTURED DATA	UNSTRUCTURED DATA
= Data that is highly organised and stored in a fixed format, usually in tables with rows and columns.	= Data that does not follow a strict table format but still has some organisation through tags or markers.	= Data that has no predefind format or structure, making it harder to store and analyse.
e.g Customer information	e.g JSON or XML files used in web applications	eg Images, videos, audio files...

6. DATA MODELING

= The process of defining how data is structured, stored, and related within a system.

* Data modeling ensures that data is organised efficiently, reducing errors and duplication. It also makes systems easier to design, maintain, and scale.

* i) Conceptual Data Model
  ii) Logical Data Model
  iii) Physical Data Model

7. DATA MINING

= The process of analysing large sets of data to discover patterns, trends, and other useful insights.

* Businesses use data mining to make better decisions by identifying customer behaviour, predicting trends, and improving operations.

* - Online stores analyse customer purchase history to suggest products they are likely to buy.
  - Banks use data mining to detect unusual transaction to detect fraud.

09/06/2026

WILD CARDS

- a way of filtering data using a specific pattern in a specific column.
- You filter data based on how words are arranged on a specific column.
  - A) % TYPE - Represents zero or more characters.
  - B) _ TYPE - Represents a single character
- In order to use wildcards we need to use the LIKE statement
- The LIKE statement is commonly used in the WHERE statement to filter data.

11/05/2026

CASE STATEMENT - to build logical condition.
- Allows us to build logical conditions.

```
SELECT
    CASE
    WHEN condition 1 THEN result 1.
    WHEN condition 2 THEN result 2
    ELSE default result
    END AS column name
FROM table name.
```

-> We write our case statement within the select statement

-> A case statement creates a new column

We build conditions based on other columns
A Case statement is a way of building a categorical column using numerical column